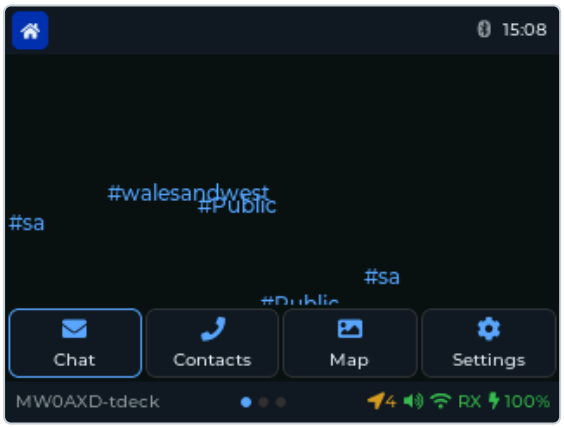
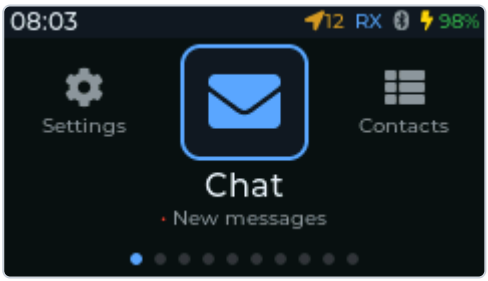
Saitama

User guide for the LilyGo T-Deck Plus
and the M5Stack Cardputer ADV

Full touchscreen on the T-Deck · Verified direct messages · Messages that get through
No phone, no internet, no subscription



T-Deck Plus



Cardputer ADV

Firmware version 1.6.5 · 9 November 2026
github.com/868meshbot/Saitama · GPL-3.0-or-later

Contents

1. Why a mesh? Why MeshCore?
2. Why Saitama?
3. Getting your message through
4. The apps explained
5. Installing Saitama
6. First setup
7. Controls
8. The launcher
9. Chat and channels
10. Contacts, Repeaters and Heard
11. Regions and scopes
12. Network tools
13. Map and GPS
14. Settings reference
15. Terminal commands
16. Backup and security
17. Troubleshooting
18. Glossary

Saitama user guide

Saitama is free, open-source firmware that turns a LoRa handheld into a standalone MeshCore messenger. You get encrypted channels and direct messages, a list of every station you hear, repeater and region tools, and (on the T-Deck) offline maps. No phone, internet or subscription is needed.

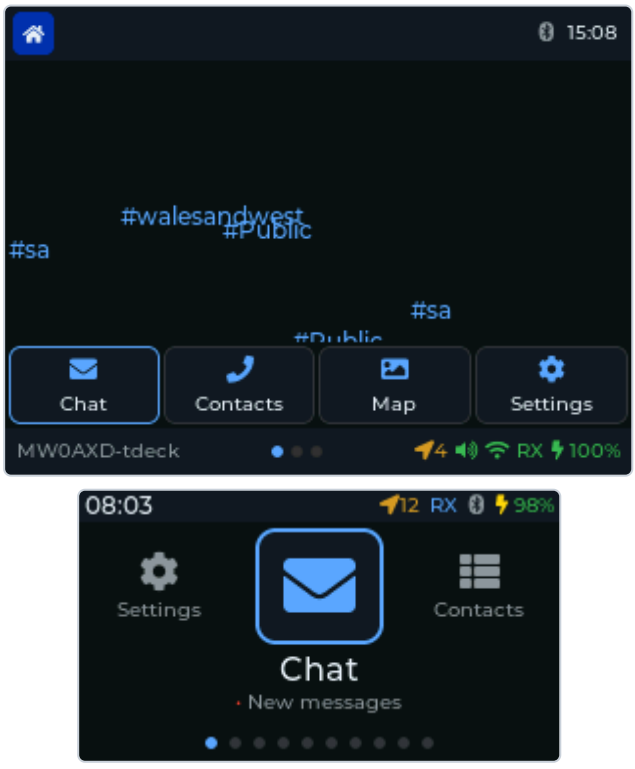
Highlights:

- **Full touchscreen on the T-Deck.** Tap, swipe and scroll on every screen, like a phone, with the keyboard and trackball still there when you want them.
- **Verified direct messages.** A green padlock shows a DM really came from that person and wasn't faked.
- **Messages that get through.** Clear delivered / not-delivered ticks, automatic resends that find a new route when the old one breaks, and routes remembered across restarts.
- **No phone, no internet, no subscription.** Everything runs on the device.

New to all this? Start with Why a mesh?, Why Saitama? and The apps explained.

This guide covers release **1.6.2** on both supported devices:

	LilyGo T-Deck Plus	M5Stack Cardputer ADV
Status	Main target, hardware-tested	Beta, hardware-tested
Screen	320 × 240, **full touch support**	240 × 135, no touch
Input	Keyboard, trackball, touchscreen	Keyboard only
Radio	Built-in SX1262	Cap LoRa-1262 module
GPS	Built-in	On the Cap LoRa-1262
Firmware files	`saitama-t-deck-plus-X.Y.Z.bin`	`saitama-cardputer-adv-X.Y.Z.bin`



The same identity, keys, contacts and settings work on both devices: the SD card's `/ops/` folder can move from one to the other.

Why a mesh? Why MeshCore?

Messages without the phone network

Your phone needs a phone mast or Wi-Fi to send anything. Usually that's fine. But there are plenty of times when it isn't:

- **Out in the hills, at sea, or deep in the countryside**, where there's no signal.
- **At a festival, a big match or a protest**, where thousands of phones jam the local mast and nothing gets through.
- **In a power cut, storm or flood**, when masts go down or run out of battery.
- **When you'd simply rather not** put every message through a phone company, an app store and somebody's servers.

A **mesh** is a way round all of that. Each device is a small radio. It talks directly to other devices nearby, with no mast, no internet and no monthly bill.

How a mesh works (the short version)

Imagine passing a note across a crowded room. You can't reach the far side, so you hand it to someone near you, who hands it on, and so on until it arrives.

A mesh does the same with radio messages:

1. You type a message. Your device sends it out by radio.
2. Any device in range hears it. Some of them are **repeaters**: small, always-on radios, often on a roof or a hill, whose job is to pass messages on.
3. Each repeater sends it on again, one **hop** further, until it reaches the person it's for.

The more people and repeaters there are, the further messages travel. Nobody owns the network: it's made of everyone's devices together.

Why LoRa?

The radios use **LoRa** ("long range"). It sends small amounts of data a long way on very little power:

- **Distance:** a few kilometres between handhelds in town is normal. From a hill or a high repeater it can be tens of kilometres.
- **Battery:** the radio sips power, so a device lasts for days, not hours.
- **No licence needed:** LoRa uses frequency bands that anyone may use without a radio licence (868 MHz in the UK and Europe, 915 MHz in the US and Australia).

The catch: LoRa is slow. It's made for text messages, not photos or video. That is exactly what you want in an emergency: short messages that get through.

Why MeshCore?

There's more than one way to run a LoRa mesh. **MeshCore** is the one Saitama uses, because it's good at the things that matter:

- **It finds a route, then uses it.** The first message to someone spreads out to find them (a **flood**). After that, MeshCore remembers the path, so later messages go straight there instead of filling the airwaves. That keeps the network quiet, so it works with many more people.
- **Repeaters are separate.** Your handheld doesn't relay everyone else's traffic, which saves your battery. Dedicated repeaters do that job.
- **It's encrypted.** Direct messages can only be read by you and the person you sent them to. Private channels can only be read by people who have the key.
- **It's open source and free**, with a growing community and repeaters already running in many areas.

Why messaging matters

When things go wrong, the first thing people need is to say "I'm OK", "where are you?" or "we need help here". A text message is the smallest, most reliable way to do that. A mesh that's already set up and already in your pocket means you have it ready before you need it.

And on a normal day it's simply fun: chatting with people nearby, seeing who's around, and being part of a network that you and your neighbours run yourselves.

Why Saitama?

Most MeshCore radios are **a radio with a phone attached**. The radio does the talking, but you read and type everything in an app on your phone, connected by Bluetooth. If your phone is flat, lost or left at home, the radio is useless.

Saitama is the whole thing in one device. It runs on handhelds that already have a screen and a keyboard, so you can read, type and send messages with nothing else.

Full touchscreen on the T-Deck

On the LilyGo T-Deck, Saitama works completely by touch, just like a phone.

Many firmwares for the T-Deck ignore its touchscreen, or only use it on a screen or two. You end up rolling a tiny trackball for everything. Saitama supports touch **everywhere**:

- **Tap** apps, buttons, list rows, contacts and messages.
- **Swipe** between the launcher pages, and scroll through chats and lists with your finger.
- **Drag** sliders and pick from drop-down lists by touch.
- Pop-up menus use **big two-column buttons** sized for fingers, not fiddly little links.
- The touch controller is found automatically at start-up, and if taps ever land in the wrong place there's a quick calibration (`touch-cali` in the Terminal).

The keyboard and trackball still work everywhere too, so use whichever you like: type with your thumbs, tap to choose, or never take your hands off the keys.

Verified direct messages: know who you're talking to

Every direct message in Saitama is checked to make sure it really came from the person whose name is on it. You can see that it's been checked: it has a green padlock 🔒 .

Why this matters: on a radio network, anyone can transmit anything. On a group channel, somebody can type any name they like and pretend to be someone else. A fake "This is Dad, meet me at the bridge" is exactly the kind of message you don't want to fall for.

A direct message is different. It's locked with a key that only the two of you share: your device and your contact's device each work it out from their own secret key and the other's public key. Nobody else has it. So:

- **It can't be read** by anyone else, and
- **it can't be faked.** Someone without your contact's secret key can't make a message that your device will accept as coming from them.

Saitama shows you the result:

- **T-Deck:** a **green padlock** beside the sender's name on each verified message.
- **Cardputer:** a **green padlock** in the title bar of the conversation.

No padlock = not verified. Channel messages never have one, because everyone in a channel shares the same key. Treat a name on a channel as "probably them", and use a DM when it matters.

The small print: your device trusts a contact's key from the first advert it saves for them, so add contacts you know are genuine. The check is MeshCore's standard message check, which stops casual forgery and tampering, not a full digital signature.

Built to get your message through

Sending a message is easy. Making sure it **arrives** is the hard part, and it's where Saitama does a lot of work you never see:

- ✓ **Delivered or ✕ Not delivered.** Every DM ends with a clear answer. Nothing vanishes silently.
- **Smart retries.** It resends along the known route; if that route has broken it forgets it and searches the whole mesh for a new one, then remembers what works. The other person still sees your message only once.
- **Routes remembered.** Proven routes are saved to the SD card, so after a restart your messages go straight to people instead of searching again.
- **"repeated ×3".** Channel messages show how many repeaters passed them on, so you know they got out.
- **Ready for big meshes.** Region tags, longer route codes for very large networks, and room made for newcomers when the device's node table fills up.

The full story is on Getting your message through.

What makes it different

No phone needed. Chat, contacts, maps, settings: everything is on the device. (You can still connect the MeshCore phone app if you want to.)

Easy to use. Big buttons, clear lists, and the same layout everywhere. On the T-Deck you tap the screen like a phone (see above). On the Cardputer everything works from the keyboard.

Free, with nothing locked away. No subscription, no "pro" version, no features held back for paying users. All of it is open source (GPL), so anyone can check what it does, fix it or improve it.

Not just messages. Saitama includes tools that usually need a laptop or a second device: see who's around, check your signal, look after repeaters, find a route, view offline maps. See The apps explained.

Your data stays yours.

- Messages are encrypted from end to end by MeshCore.
- **Direct messages are verified**, with a green padlock to prove it (see above).
- Keys and passwords saved to the SD card are locked with your own password.

Survives a reflash. Your identity, contacts, repeaters and settings are backed up to the SD card. Update the firmware or swap to another device, and everything comes back.

Works with everyone else. Saitama speaks standard MeshCore, so you can message people using any other MeshCore device or app.

Saitama compared

How Saitama compares with the usual way of using MeshCore: a radio paired with the phone app.

	Typical MeshCore radio + phone app	Saitama
Works without a phone	No: the radio has no screen or keyboard of its own	**Yes: everything is on the device**
Touchscreen	On the phone	**Full touch on the T-Deck itself**
Phone battery	Drained by the app and Bluetooth	Not needed at all
DM resends	Up to the phone app; if the phone is off, nothing happens	**Built into the device: route first, then a fresh search**
Repeater admin	In the phone app	**On the device, with a remembered password**
Offline maps	On the phone, if you've downloaded them	**On the device, from the SD card (T-Deck)**
Cost	Free	**Free and open source, nothing locked behind a paywall**

You can still pair the MeshCore phone app with Saitama over Bluetooth if you'd like to.

Which device should I get?

	LilyGo T-Deck Plus	M5Stack Cardputer ADV
In short	The full experience, with touch	Small, cheap and pocketable
Screen	320 × 240, touch	240 × 135, no touch
Touchscreen	**Yes: full touch support on every screen**	No
Typing	Keyboard + trackball	Small keyboard
GPS	Built in	On the LoRa module
Apps	All of them, including maps, radio tools, music and games	The essentials: chat, contacts, repeaters, finder, signal, GPS
Status	Main device, well tested	Beta, tested

If you want a touchscreen that works like a phone, maps or the radio tools, get the T-Deck. If you want something pocket-sized for messaging and checking the mesh, the Cardputer does the job.

Getting your message through

Radio is messy. A message can be lost because a repeater was busy, someone walked behind a hill, or two radios talked at the same moment. A good mesh device doesn't pretend that never happens: it notices, tries again, and tells you what happened.

This is where Saitama puts a lot of its effort.

You always know if a DM arrived

Every direct message you send ends in one of two ways, shown on your message bubble:

- ✓ **Delivered.** The other person's device confirmed it got the message.
- ✕ **Not delivered.** Saitama tried everything it could and got no confirmation.

There's no "sent… and who knows?" A message never just quietly disappears.

It tries again for you, and changes its approach

If a DM isn't confirmed in time, Saitama doesn't simply send the same thing again and hope. It works through a plan:

1. **Try the known route again.** If it knows a path to the person, it resends along that path, up to **5 times**.
2. **Give up on a bad route.** If the route keeps failing, the route is probably broken (a repeater is down, or the person has moved). Saitama **forgets it**, both in memory and on the SD card, so it won't reuse it after a restart.
3. **Search the whole mesh.** It then sends the message as a **flood**, which spreads out through every repeater to find the person wherever they are now, up to **3 times**.
4. **Learn the new route.** When the reply comes back, the new path is remembered for next time.

You can change the number of tries in **Settings > DM Retries**.

Clever details that make retries work

- **Repeaters don't ignore the retries.** Repeaters skip messages they've already passed on. Saitama makes each retry a slightly different packet, so repeaters forward it as new.
- **No double messages.** The person you're messaging still only sees your message **once**, however many retries it took.
- **Several at once.** Up to 4 DMs can be on their way at the same time, each with its own retries and timer. A slow one doesn't hold up the rest.

Routes are remembered, and kept fresh

The first DM to someone has to search for them. After that, Saitama remembers the route so later messages go straight there: faster, and less traffic for everyone.

- **Only proven routes are kept.** A route is saved after a message along it is confirmed, with the time it was confirmed. Guesses aren't saved.

- **Routes survive a restart.** They're saved with your contacts and repeaters on the SD card, and loaded back when the device starts.
- **Repeaters are ready at once.** Saved repeaters are loaded at start-up, so you can log in to one straight away, without waiting to hear from it again.
- **You're in control when you need to be. Show Path** on a message shows the route it took. In Contacts and Repeaters you can **Set Path** by hand or **Reset Path** to start fresh, and on the T-Deck **Trace** shows the signal at every hop.

See that your channel message got out

Channel messages go to everyone, so nobody sends back a "got it". Instead, Saitama listens for 30 seconds after you send and counts the repeaters it hears passing your message on. Your bubble shows **repeated ×3** (for example), so you know it reached the network and how widely. Select the bubble to see which repeaters picked it up.

Built for big, busy meshes

- **Region scopes.** On meshes where repeaters only pass on messages for their own area, Saitama tags your messages correctly, and the **Regions** app finds out which tags your local repeaters use. See Regions and scopes.
- **Fewer mix-ups on large networks.** Routes name each repeater by a short code. On a mesh with hundreds of repeaters, two can share a 1-byte code. **Settings > Radio Mode > Routing** can use 2- or 3-byte codes instead.
- **New people aren't locked out.** When the device's table of known nodes is full, Saitama makes room by dropping the least recently heard non-favourite, rather than refusing newcomers (who then couldn't message you).

A device you can rely on

Getting a message through is no use if the device has crashed or lost your data.

- **Safe saving.** Contacts and repeaters are written to a temporary file first and only then swapped in, with the previous copy kept as a backup. A flat battery mid-save doesn't wipe your address book.
- **Low-memory protection (Cardputer).** The Cardputer has very little memory. If it runs short, Saitama closes the screens you aren't using and carries on, instead of crashing.
- **Everything backed up.** Your identity, contacts, routes and settings are on the SD card and come back after an update or on a new device.

The apps explained

Every app in Saitama, in plain English: what it's for and why you'd want it.

Both = on the T-Deck and the Cardputer. **T-Deck** = T-Deck only.

Talking to people

Chat (both)

What it is: your messages, like a text-message app on a phone.

Why you'd use it: to talk. Group chats are called **channels**. Everyone can read **Public**; you can make private channels for your family, friends or club. You can also send a **direct message** to one person only. Direct messages are **verified**: a green padlock shows the message really came from that person and wasn't faked.

Example: you're walking in the hills with friends, out of phone signal. You make a channel called "Walk", and everyone can see "I'm at the car park" and "Waiting at the top".

Contacts (both)

What it is: your address book of people on the mesh.

Why you'd use it: to message someone directly, star your favourites, see when you last heard from them, and block anyone causing trouble. It's also how you share your own details by QR code.

Heard (both)

What it is: a live list of every device your radio has picked up since you switched it on, with how strong their signal was.

Why you'd use it: to see who's around right now. If you've just arrived somewhere, Heard tells you at a glance whether anyone's in range. You can save anyone in it as a contact.

Looking after the network

Repeaters (both)

What it is: a list of the repeaters near you. Repeaters are the always-on radios that pass messages along.

Why you'd use it: to see which repeaters you can reach and how good the link is. If you run a repeater, you can log in to it from here: check its battery, see who it can hear, change its password, all without climbing up to it with a laptop.

Finder (both)

What it is: a quick "who can hear me right now?" scan. It only counts devices that hear you **directly**, with no repeater in between.

Why you'd use it: to find a good spot. Move around, run Finder again, and see if more devices hear you. Great for choosing where to put a repeater, or checking your antenna is working.

Regions (both)

What it is: asks nearby repeaters which **regions** they cover, and lets you save them.

Why you'd use it: some busy meshes only pass on messages marked for their area (to stop messages flooding the whole country). If your messages aren't getting out, Regions tells you which area tag to use. See Regions and scopes.

Signal (both)

What it is: the "health check" for your radio: how strong the last signal was, how noisy the airwaves are, and how many messages have gone in and out.

Why you'd use it: when something seems wrong. If the received count isn't going up, you're not hearing anything. If the noise is high, something nearby is interfering.

Advertise (T-Deck) / Settings > Advert (Cardputer)

What it is: sends out a short "Hello, I'm here" announcement, called an **advert**.

Why you'd use it: other people can only message you once they've heard your advert. Send one when you arrive somewhere new. You can also have it sent automatically every few hours.

Trace (T-Deck)

What it is: follows the path a message takes, hop by hop, and shows the signal at each step.

Why you'd use it: to find the weak link. If messages to someone keep failing, Trace shows which repeater on the way is struggling.

Where am I?

Map (both)

What it is: a proper map, stored on the SD card, so it works with no internet. People who share their location appear on it. The Cardputer uses its own JPEG tiles: see Map and GPS.

Why you'd use it: to see where your friends and nearby repeaters are, or just to find your way when your phone has no signal.

GPS (both)

What it is: shows your position, the time, and how many satellites the GPS can see. On the T-Deck it draws a sky map of the satellites.

Why you'd use it: to check you've got a fix before sharing your location, or to read your exact coordinates out to someone.

Tools for the curious (T-Deck)

These are for people who like to see what's happening on the airwaves. You don't need any of them for everyday messaging.

App	In plain English	Why it's useful
ChanScan	Listens on the channels next to yours and shows how busy each one is.	Spot interference, or find a quieter frequency.
Spectrum	A picture of radio activity across the band, like a graphic equaliser.	See signals and interference you can't hear.
SigGen	Sends a test signal.	Testing antennas and other radios.
PCAP	Records the raw messages your radio hears to the SD card.	For experts studying how the network behaves.
BT Foxhunt	Beeps faster as you get closer to a chosen Bluetooth device.	A "hot and cold" game for finding a lost tracker, or a fun group hunt.
Power	Battery history and how long it should last.	Know whether you'll make it through the day.

Everything else

App	Device	What it's for
Settings	Both	Your name, radio, sounds, screen, passwords: everything you can change. See Settings reference.
Terminal	Both	Type commands, like an old-school computer. Handy for power users and for fixing problems. See Terminal commands.
MP3	T-Deck	Plays music from the SD card.
Pic Viewer	T-Deck	Shows pictures from the SD card.
Files	T-Deck	Look through and tidy the SD card.
Snake, **2048**	T-Deck	Games, for when you're waiting for a reply.

Installing Saitama

What you need

- **T-Deck:** a LilyGo T-Deck Plus (the plain T-Deck should work but is untested).
- **Cardputer:** an M5Stack Cardputer ADV with the Cap LoRa-1262 module fitted.
- A USB-C data cable (some cables only charge).
- A microSD card, FAT32. Optional, but strongly recommended: it holds your message history and the backup of your identity, contacts and settings.
- An antenna on the radio before you send anything. Transmitting without one can damage the radio.

Download

Get the files from the latest release. Each release has two files per device:

File	Flash at	Use it for
`saitama-<device>-X.Y.Z-merged.bin`	`0x0`	First install, or recovery. Bootloader, partition table and app in one file.
`saitama-<device>-X.Y.Z.bin`	`0x10000`	Updating a device that already runs Saitama, or installing from M5Launcher.

`<device>` is `t-deck-plus` or `cardputer-adv`. Don't mix them up: each file only runs on its own device.

Flashing tools

Any ESP32-S3 flasher works. Two common ones:

- **In a browser:** ESP Tool in Chrome or Edge. Connect, add the file at the address from the table above, and press Program.
- **From a terminal:** esptool (`pip install esptool`).

T-Deck Plus

1. Put the T-Deck into download mode: hold the trackball button down, press the reset button on the side, then let go of both.
2. Connect it by USB.
3. Flash the merged file at `0x0` (command below).
4. Press reset. The Saitama splash appears, then the home screen.

```
esptool.py --chip esp32s3 --port <port> write_flash 0x0 saitama-t-deck-plus-1.6.2-merged.bin
```

`<port>` is something like `/dev/ttyACM0` (Linux), `/dev/cu.usbmodem…` (macOS) or `COM5` (Windows).

To update later, flash the app-only `.bin` at `0x10000` the same way.

Cardputer ADV

With a USB flasher:

1. Hold the **G0** button while plugging in USB. This starts download mode.
2. Flash the merged file at `0x0` (command below; the Cardputer ADV has 8 MB of flash).
3. Unplug and plug back in (or press reset).

```
esptool.py --chip esp32s3 --port <port> write_flash 0x0 saitama-cardputer-adv-1.6.2-merged.bin
```

With M5Launcher: copy `saitama-cardputer-adv-X.Y.Z.bin` (the app-only file) to the SD card and install it from M5Launcher's file browser.

Updating

- Flashing the **app-only** file keeps everything on the device: settings, identity, contacts.
- Flashing the **merged** file can reset the settings held on the device. With an SD card in, Saitama restores them from the `/ops/` backup on the card at the next boot.

Whichever you use, it's worth keeping the SD card in. See Backup and security.

Building it yourself

```
git clone --recurse-submodules https://github.com/868meshbot/Saitama.git
cd Saitama
pio run -e t-deck -t upload          # T-Deck / T-Deck Plus
pio run -e cardputer-adv -t upload   # Cardputer ADV
```

This needs PlatformIO.

First setup

Everything below is in **Settings**. Do these once after the first install.

1. Pick your radio preset

Settings > Radio. Every node on a mesh must use the same frequency, spreading factor, bandwidth and coding rate, or they can't hear each other. Choose the preset your local mesh uses:

Preset	Settings	Frequency
Australia	SF10 BW250 CR5	915.800 MHz
Australia Vic.	SF7 BW62 CR8	916.575 MHz
EU/UK Narrow (Rec) (default)	SF8 BW62 CR8	869.618 MHz
EU/UK Long Range	SF11 BW250 CR5	869.525 MHz
EU/UK Medium	SF10 BW250 CR5	869.525 MHz
Czech Narrow	SF7 BW62 CR5	869.525 MHz
EU 433 Long Range	SF11 BW250 CR5	433.650 MHz
New Zealand	SF11 BW250 CR5	917.375 MHz
NZ Narrow	SF7 BW62 CR5	917.375 MHz
Portugal 433	SF9 BW62 CR6	433.375 MHz
Portugal 868	SF7 BW62 CR6	869.618 MHz
Switzerland	SF8 BW62 CR8	869.618 MHz
USA/Canada	SF7 BW62 CR5	910.525 MHz
Vietnam	SF11 BW250 CR5	920.250 MHz
Japan	SF12 BW125 CR8	920.800 MHz
Custom	your own values	

Only use frequencies that are legal where you are. **Japan** (ARIB STD-T108): transmit power is capped at 13 dBm (20 mW) while this preset is selected, and frequency hopping is not available. The rules there also require listen-before-talk and a 4 s limit per transmission, which Saitama does not enforce yet.

2. Set your name

Settings > Device Name. This is the name other people see on your messages and adverts. It starts as `OMS-0001`; change it to something people will recognise.

3. Change the storage password

Settings > Storage Password. The row says **Default** in red until you change it.

Channel keys, remembered repeater passwords and your identity backup are encrypted on the SD card with this password. The default (`changeme1`) is published, so until you change it, anyone who takes your

SD card can read them. Pick a password you'll remember: you need it to restore your identity onto a new or reflashed device. See Backup and security.

4. Set the clock

Settings > Timezone, then **Settings > Date / Time**. With a GPS fix the time is set from GPS automatically. The clock is used for message times and Night Mode.

5. Tell the mesh you're here

Other nodes learn about you from your **advert**. Until they've heard one, they can't send you a direct message.

- **T-Deck:** open the **Advertise** app and send one.
- **Cardputer: Settings > Advert** sets how often an advert goes out on its own (every 12 hours by default). To send one now, type `advert` in the Terminal.

Optional

- **Notifications** and **Notification Sound**: popups and sounds for new messages.
- **Night Mode**: quiet hours, with sounds muted.
- **Bluetooth** and **Bluetooth PIN**: to use the MeshCore phone app with the device.
- **Screen Timeout**: how long before the screensaver starts (10 s by default).
- **Theme**, **Font**, **Brightness**, **Language**.
- **Telemetry**: whether others may ask for your battery level and position.

All of these are explained in the Settings reference.

Controls

T-Deck Plus

The T-Deck can be used entirely by touch, entirely by keyboard and trackball, or by any mix of the two.

To do this	Touch	Keyboard / trackball
Move between items		Roll the trackball
Select / open	Tap	Click the trackball, or Enter
Go back, close a popup	Tap the **X** (top right) or **Close**	Backspace
Scroll a list or chat	Swipe up or down	Roll the trackball
Change launcher page	Swipe left or right	Roll the trackball past the edge
Type		Keyboard
Wake the screensaver	Tap	Any key, or the trackball

If taps land in the wrong place, calibrate the screen: open the **Terminal** and type `touch-cali`, then tap each crosshair in turn. `touch-cali reset` undoes it.

The keyboard backlight is set in **Settings > Keyboard Light** (a brightness level, Off, or Auto, which lights it between set times and during Night Mode).

Cardputer ADV

The Cardputer has no touchscreen: everything is done from the keyboard. The highlighted item has an outline in the accent colour.

To do this	Keys
Move up / down / left / right	`;` `.` `,` `/` (the keys with arrows printed on them)
Move between fields in a form	Tab
Select / open / tick a box	Enter, or the **G0** button
Go back, close a popup	Backspace, or Fn + `` ` `` (Esc)
Close a QR code	Any key
Move inside a text box	Hold **Fn** with the arrow keys
Wake the screensaver	Backspace or Esc
Wake a blank screen	Any key
Map: move / zoom / centre on you	Arrows / `=` `-` / `c`

In a text box the arrow keys type their characters (`;` `.` `,` `/`). Press Tab or Enter to leave the box, or hold Fn to use them as arrows.

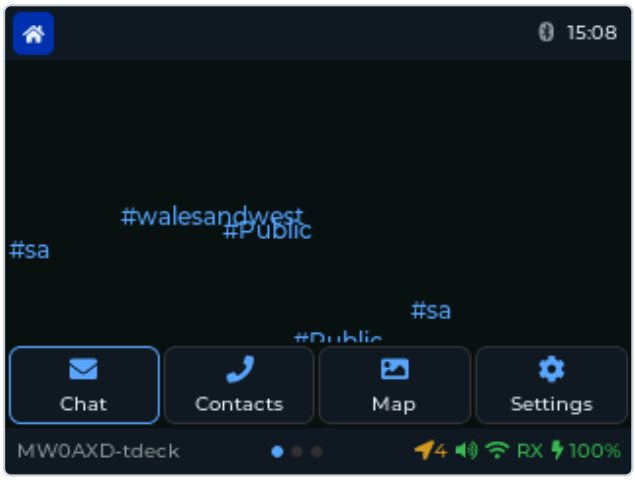
Drop-down lists open as a full-screen column of buttons: move to the option you want and press Enter, or press Backspace to leave it unchanged.

The launcher

T-Deck Plus

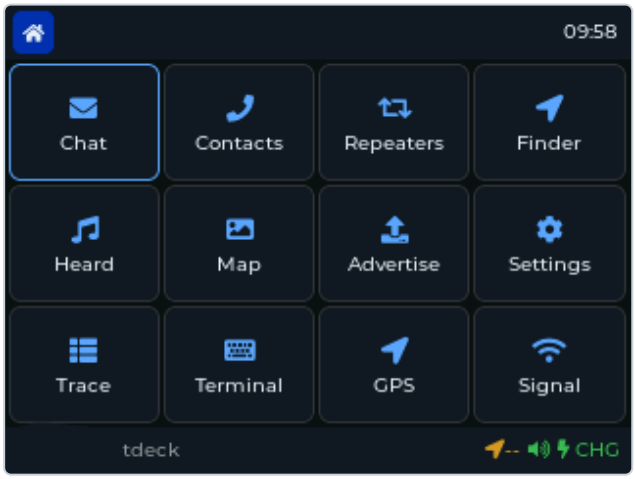
The launcher has three pages. Swipe left or right, or roll the trackball past the edge of a page, to change page.

Home page. The stations you've heard this session fall like rain in the background, coloured by signal strength, with channels in blue. When a message arrives, a line joins it to its channel, the station it replies to, or the DM recipient. In front are four big buttons: **Chat**, **Contacts**, **Map** and **Settings**. The clock and battery are at the top; a moon beside the clock means Night Mode is on.



Apps page 1: Chat, Contacts, Repeaters, Finder, Heard, Map, Advertise, Settings, ChanScan, Terminal, GPS, Signal.

Apps page 2: MP3, Files, Spectrum, Trace, SigGen, Power, Snake, 2048, PCAP, BT Foxhunt, Pic Viewer, Regions.

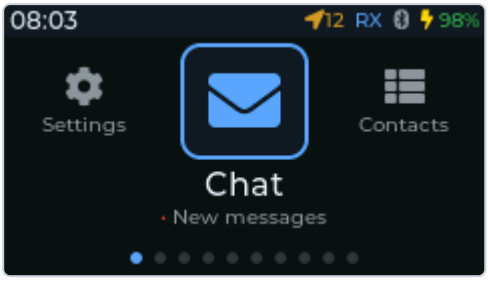


Cardputer ADV

The Cardputer has fewer apps (it has no PSRAM, so the radio tools, games and media apps are left out, and the map is a lighter version):

Chat, Contacts, Repeaters, Heard, Finder, Regions, Signal, GPS, Map, Terminal, Settings.

Carousel (the default): one big icon at a time, with its neighbours at the sides. Left and Right move round it, Enter opens the app. The line under the icon shows something live for that app: unread messages, the number of contacts or repeaters, the GPS fix, and so on.



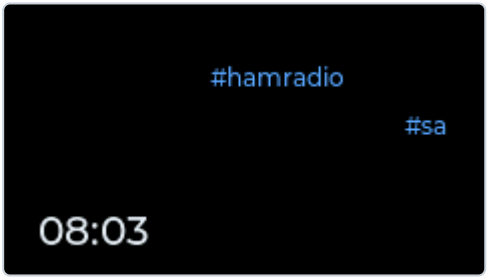
Menu: Settings > Launcher switches between Carousel and Menu, a plain list of the apps.

The Cardputer has no Advertise app: adverts are in **Settings > Advert**, and the Terminal `advert` command sends one straight away.

Screensaver

After the **Screen Timeout** (Settings) the screensaver starts. **Settings > Screensaver** picks what it shows:

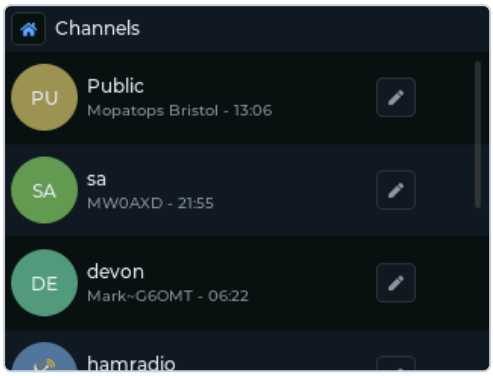
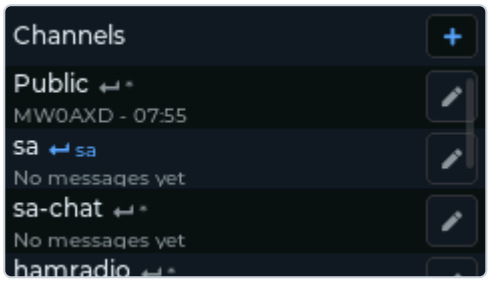
- **T-Deck:** a clock, or **Saitama** (an animated mesh).
- **Cardputer:** a clock, or **Mesh rain** (the default): channels, stations and the clock fall down the screen, joined by a line when a message arrives.



Settings > Screen Off turns the backlight off after a further delay. The radio keeps listening, and messages still arrive.

Chat and channels

Open **Chat** from the launcher. The first screen lists your channels, with an unread count and a preview of the last message, and a **Direct Messages** entry at the bottom.

T-Deck	Cardputer
	

Channels

A channel is a group chat. Everyone who has the channel's key can read it and post to it.

- **Public** is always there. Every MeshCore node has its key, so treat it as open.
- You can add up to **9 more** channels.

Adding a channel

Choose **+ Add Channel** at the top of the channel list (or use **Settings > Channels**). Fill in:

- **Name.** A name starting with `#` (for example `#Mesh`) is a **hashtag channel**: leave the key empty and it is worked out from the name, so anyone who types the same name joins the same channel. Any other name (for example `MyGroup`) is a **private channel**: leave the key empty and a random key is made for you. Share it with the people you want in.
- **Key (PSK).** To join a private channel someone else made, paste their key here.

To see a channel's key so you can share it, type `channel <n>` in the Terminal. `n` is the slot, 1 to 10 (1 is Public); `channels` lists them. It prints the key in base64.

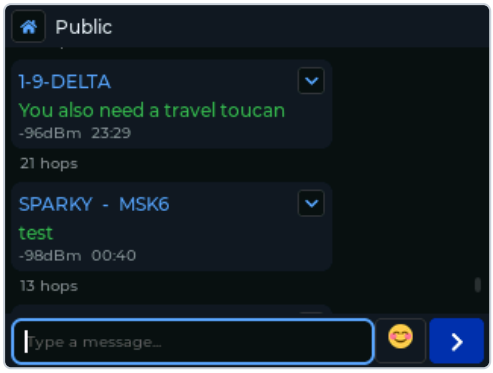
Channel options

The ≡ button at the end of a channel's row opens its options:

- **Configure Scope**: which region this channel's messages are flooded to. See Regions and scopes.
- **Channel Icon** (Cardputer): initials or an emoji in front of the channel name. The T-Deck shows each channel's initials in a circle instead.
- **Clear Messages**: empties the conversation.
- **Notify: On / Off**: popups and sounds for this channel.
- **Delete Channel** (not for Public).

Reading and sending

Open a channel to see its conversation as speech bubbles: yours on the right, others on the left with the sender's name. Type a message and press Enter to send.

T-Deck	Cardputer
	*(screenshot to come)*

Under messages you sent to a channel, **repeated ×N** shows how many times repeaters were heard passing it on: a sign it got out.

Select a message bubble to open its menu (on the T-Deck tap the bubble's menu button; on the Cardputer move to the bubble and press Enter):

- **Reply**: starts your message with `@[Name]`.
- **Show Path**: the repeaters the message came through.
- **Add Contact**: saves the sender as a contact, if their advert has been heard.

With **Settings > Show Hops** and **Show RSSI** on, each received message also shows how many hops it took and its signal strength.

Direct messages (DMs)

A DM goes to one contact only, encrypted so that only the two of you can read it.

- **Start one:** open **Contacts**, select the person and choose **Direct Message**, or open **Direct Messages** in Chat.
- **Delivery:** a tick appears on your bubble when the recipient's device confirms it. If no confirmation comes back, Saitama resends it: first along the known route, then as a flood (**Settings > DM Retries**, 5 and 3 by default). If every try fails the bubble shows ✕ **Not delivered**. See Getting your message through.
- **Green padlock = verified.** See below.

Verified DMs (the green padlock)

Every DM is encrypted and checked with a key that only you and that contact can work out: each side combines its own private key with the other's public key. A message that wasn't made with your contact's private key fails the check, so a DM **can't be forged in their name**.

- **T-Deck:** a green padlock beside the sender's name on each verified message.
- **Cardputer:** a green padlock in the conversation's title bar.

Channel messages have no padlock and **can't** be verified: anyone with the channel key can post under any name. When it matters who sent something, use a DM.

The details: the contact's public key is trusted from the first advert you save for them, and the check is MeshCore's 2-byte message authentication code (MAC), not a full digital signature.

You can only DM someone after you've heard their advert (or saved them from a QR code), and they can only DM you after hearing yours.

History

With **Settings > Save Messages** on (the default), every conversation is saved to the SD card in `/ops/msgs/` and reloaded when you open it. The Cardputer keeps fewer messages in memory per conversation than the T-Deck (10 rather than 20); older ones stay on the card.

Notifications

Settings > Notifications turns the new-message popup on or off. **Settings > Notification Sound** picks the sound (Default, Pluck, Clear or Whoosh) and **Settings > Speaker** sets the volume. Sounds are muted during Night Mode.

Contacts, Repeaters and Heard

MeshCore nodes announce themselves with **adverts**. Saitama sorts what it hears into three apps:

- **Heard**: every station heard since the device was switched on.
- **Contacts**: people (companion nodes) you've saved. Up to 250 on the T-Deck, 100 on the Cardputer.
- **Repeaters**: repeaters you've saved. Up to 250 on the T-Deck, 60 on the Cardputer.

Contacts and repeaters are saved to the SD card (`/ops/contacts.json` and `/ops/repeaters.json`) and survive a reflash.

Heard

A live list of stations, newest first. Each row shows **C** (companion) or **R** (repeater), the name, signal bars for the last packet heard, and (T-Deck only) the time it was last heard.

Select a row for its details: name, identity, signal and, for a repeater, its owner information (asked for over the air; it can take a few seconds). **Save** adds it to Contacts or Repeaters; **Update** refreshes a saved one. **Clear List** (top right) empties the list.

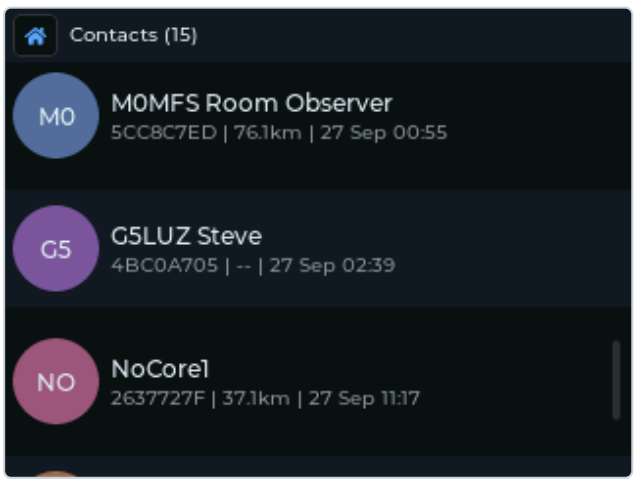
Adding contacts

Any of these work:

- **Automatically: Settings > Auto Add Contacts** saves nodes as their adverts arrive.
- **From Heard:** select the station and choose **Save**.
- **From a message:** select the bubble in Chat and choose **Add Contact**.
- **QR code: Settings > Share My Contact** shows your own QR code for another device or the MeshCore phone app to scan.
- **Terminal:** `card` prints your business card; `import <card>` adds someone else's.

Contacts

The list shows a star beside favourites, signal bars, and a speech-bubble button that opens a DM straight away. Type to filter the list by name.



Select a contact for its options:

Direct Message	
Share QR	**Details**
Favourite / **Remove Favourite**	
Set Path	**Reset Path**
Delete Contact	**Block Contact**

- **Details:** ID, when last heard, signal, hop count, route, distance and position (if they share it), and their public key.
- **Set Path / Reset Path:** fix the route DMs take, or forget it so the next DM floods and a new route is learned.
- **Block Contact:** their DMs and channel messages are dropped and no longer shown.

Repeaters

The same layout as Contacts. Select a repeater for:

Admin Login	
Add / Remove Favourite	
Set Path	**Reset Path**
Trace Repeater (T-Deck)	**Delete Repeater**

Repeater admin

If you know a repeater's admin password you can manage it from the device.

1. Choose **Admin Login** and type the password.
2. Tick **Remember password** to keep it for next time (sealed with your storage password on the SD card). On the Cardputer, press Down to reach the tickbox and Enter to tick it.
3. Press Enter, or choose **Login**.

If the login fails you can **Retry**, or **Reset & Retry**, which forgets the stored route so the login floods across the mesh.

Once logged in, the admin panel has:

- **Status:** battery, radio, packet counts, airtime, uptime and any sensor readings.
- **Neighbours:** the repeaters it can hear, with names.
- **Change Password**, **Access Control** (who may log in), and **Regions** (which region scopes it serves).

Any other repeater command can be sent from the Terminal with `repadmin <cmd>`.

Regions and scopes

Some meshes set their repeaters to forward only traffic tagged with a **region scope**, such as `#london`. A message without the right tag isn't repeated, so it only reaches people in direct radio range. If your messages don't seem to get out, ask locally whether your mesh uses scopes.

Your default scope

Settings > Region Scope has 10 slots:

- Slot 1 is your **default scope**, used for every flood you send (adverts, channel messages, DMs that flood). `*` means no scope.
- Slots 2 to 10 are **saved regions** you can pick from elsewhere.

Saitama never adds a region on its own: you choose them.

Per-channel scope

A channel can use a different scope from the default. In Chat, press the ≡ button on the channel's row and choose **Configure Scope**.

The Regions app

Regions asks the repeaters you can hear directly (zero hops) which regions they serve, and lists the answers. Select a region to save it to one of your slots. Your saved regions are shown as chips under the list.

On the T-Deck, Regions is on the second app page.

From the Terminal

- `scope` shows the current default scope.
- `scope #<region>` sets it, for example `scope #london`.
- `scope clear` removes it.

Network tools

On both devices

Finder. A zero-hop scan: it asks which nodes can hear you directly, without going through a repeater, and lists them with their signal. Good for checking your antenna and position, or finding the nearest repeater.

Signal. Live radio figures: last RSSI and SNR, the noise floor, packets sent and received (flood and direct), errors, and airtime used for sending and receiving.

Advert. Tells nearby nodes you exist (see First setup).

- T-Deck: the **Advertise** app.
- Cardputer: **Settings > Advert** for the automatic interval (Off, 3, 6, 12 or 24 hours), or `advert` in the Terminal to send one now.

Sending any advert, by hand or automatically, restarts the countdown to the next automatic one.

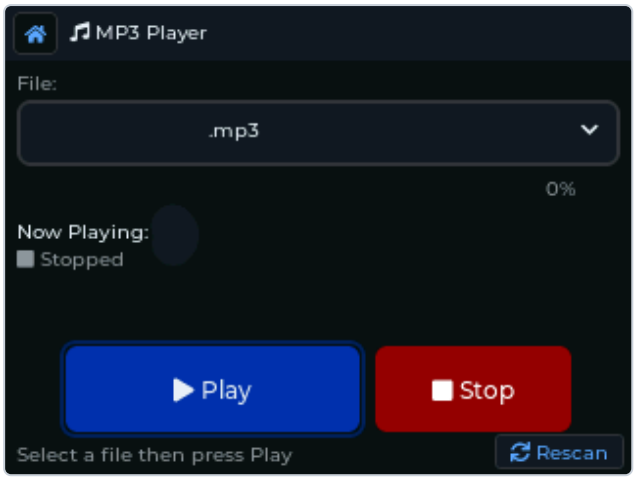
T-Deck only

The Cardputer has no PSRAM, so these are left out of its build.

App	What it does
Trace	Follows a route hop by hop and shows the signal at each repeater. Also opened by **Trace Repeater** in the Repeaters app.
ChanScan	Checks the channels either side of the mesh frequency for LoRa activity, and shows how busy each one is.
Spectrum	A spectrum and waterfall view of the band.
SigGen	A signal generator, for testing.
PCAP	Captures received packets to the SD card for later analysis.
BT Foxhunt	Tracks a Bluetooth device by signal strength, beeping faster as you get closer.
Power	Battery history and estimated running time.

Media and extras (T-Deck only)

MP3 player (from the SD card), **Pic Viewer**, **Files** (file manager), and the games **Snake** and **2048**.


MP3 Player
File:
.mp3
0%
Now Playing:
Stopped
Play
Stop
Select a file then press Play
Rescan

Map and GPS

GPS

Settings > GPS sets the receiver to **Off**, **Intermittent** (wakes now and then to save battery) or **On**. The GPS needs a view of the sky for its first fix, which can take a few minutes.

With a fix, the clock is set from GPS, and your position can be shared:

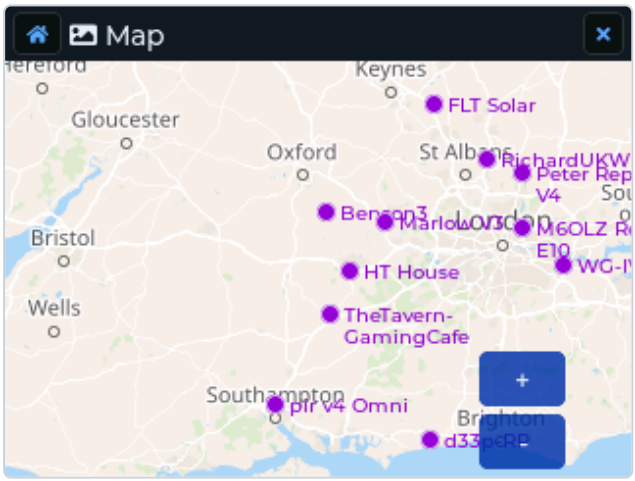
- **Settings > Location Sharing**: include your position in your adverts.
- **Settings > Telemetry**: whether others may ask your device for battery level and position.

The **GPS** app shows the fix:

- **T-Deck:** a sky plot of the satellites in view, with position, altitude and time.
- **Cardputer:** the same information as text.

Offline map (T-Deck)

The **Map** app shows map tiles from the SD card, with the positions of nodes that share their location.



Putting tiles on the card

Tiles go on the FAT32 SD card, one folder per style, in the usual `{zoom}/{x}/{y}.png` layout:

```
/maps/osm/day/{z}/{x}/{y}.png      standard OpenStreetMap
/maps/osm/night/{z}/{x}/{y}.png    optional dark style
```

With the included script (from a copy of the Saitama repository):

```
python3 scripts/download_tiles.py \
    --output /Volumes/SD/maps/osm \
    --lat 51.5 --lng -0.1 --radius 20 --zoom 10-14
```

Use your own latitude and longitude. `--radius` is in kilometres. Stay within the OpenStreetMap tile usage policy: the script is limited to 2 requests a second, and large areas at high zoom are a lot of tiles.

With a GUI tool: map-tiles-downloader. Export standard OSM tiles and put them under `/maps/osm/day`.

Night tiles

Settings > Map Style is **Day**, **Night** or **Auto** (the default: night tiles while Night Mode is on). Without night tiles on the card, the day tiles are shown colour-inverted at night, so no extra download is needed.

Offline map on the Cardputer

The Cardputer has its own, lighter map. It reads small **JPEG** tiles from the SD card (the T-Deck's PNG tiles are too big to decode without PSRAM) and draws them straight to the screen. It shows repeaters (purple), contacts (red) and you (blue), with the zoom level and the map centre on the bottom line. There is no night style.

Key	Does
`;` `.` `,` `/` (arrows)	Move the map
`=` / `-`	Zoom in / out
`c`	Centre on your position
Backspace or Esc	Back to the launcher

With no GPS fix and no manual position, the map opens on the tiles that are on the card, and zoom stops at the levels the card has.

Putting tiles on the card

Tiles go on the SD card as:

```
/maps/osm/{z}/{x}/{y}.jpg
```

They must be ordinary (baseline) RGB JPEGs, 256 x 256. The included script `scripts/map_tiles_cardputer.py` (needs Python 3 and Pillow: `pip install pillow`) makes them in a local `sdcard/maps` folder; copy that `maps` folder to the root of the card.

You need a tile provider that allows offline downloads. OpenStreetMap's own server does not: it answers bulk downloads with an "Access blocked" picture, and the script stops when it sees one. We use **Thunderforest**, which needs an **API key**, and permits bulk/offline downloading only on its Small Business plan or higher. Create an account, pick the plan, and copy the API key from your Thunderforest dashboard. Keep the key private: don't put it in a file you share or commit.

Download an area around a point (25 km radius by default, zoom 8 to 14):

```
python3 scripts/map_tiles_cardputer.py --lat 51.50 --lon -3.70 \
    --url "https://tile.thunderforest.com/atlas/{z}/{x}/{y}.png?apikey=YOUR_KEY"
```

Or a box, south west north east:

```
python3 scripts/map_tiles_cardputer.py --bounds 49.9 -5.8 52.0 0.5 \
    --url "https://tile.thunderforest.com/atlas/{z}/{x}/{y}.png?apikey=YOUR_KEY"
```

`atlas` is a clean style that reads well on the small screen; `outdoors` (paths and contours) also works. Add `--dry-run` first to see how many tiles an area is: every tile counts against your Thunderforest quota. Southern England and Wales at zoom 8 to 14 is about 59,000 tiles and 125 MB of data (more on the card, as each file takes at least one cluster). Run the script again to fill any gaps: tiles already there are skipped.

Already have T-Deck tiles? Convert them with no download:

```
python3 scripts/map_tiles_cardputer.py --from /Volumes/SD/maps/osm/day
```

Other options: `--radius-km`, `--min-zoom`, `--max-zoom`, `--quality` (JPEG quality, default 60), `--out` (output folder) and `--delay` (seconds between downloads).

Settings reference

Rows are listed in the order they appear in **Settings**. A row's value is shown on the right; green means on, red means off or needing attention. Rows marked T-Deck or Cardputer only appear on that device.

Identity and radio

Setting	What it does
Device Name	The name others see. Default `OMS-0001`.
Share My Contact	Shows your contact as a QR code for another device or the MeshCore app to scan.
Generate Identity	Makes a new key pair, so you become a new node on the mesh. Your contacts would need to add you again. Rarely needed.
Channels	Add, edit or remove channels and their keys (PSK). Empty key on a `#name` channel = derived from the name.
Radio	The radio preset (see First setup) or Custom frequency / SF / BW / CR.
Power	Transmit power.
Radio Mode	**Continuous RX** (always listening), **LoRa Duty Cycle** (the radio sleeps 250 ms in every 500 ms when idle: saves power, adds a little delay), or **FHSS** (frequency hopping, only on 915 MHz presets; see `docs/FHSS.md`). Also holds **Routing**: 1, 2 or 3-byte path hashes.
CPU Governor	Processor speed against battery life.

Display and sound

Setting	What it does
Brightness	Screen backlight.
Theme	Colour scheme: Default, Green, Dracula, Tokyo Night, Catppuccin, Synthwave 84 and more.
Font	The font used for messages and names.
Speaker	Speaker on/off and volume.
Night Mode	Start and end times. During Night Mode, sounds are muted and a moon shows by the clock.
Map Style (T-Deck)	Day, Night or Auto. See Map and GPS.
Launcher (Cardputer)	Carousel or Menu.
Screen Timeout	Seconds of no use before the screensaver starts (default 10).
Screen Off	Turns the backlight off after the screensaver has run this long.
Screensaver	T-Deck: Clock or Saitama. Cardputer: Clock or Mesh rain.
Notifications	New-message popup on or off.

Setting	What it does
Notification Sound	Default, Pluck, Clear or Whoosh.
Language	English, Italiano, Français, Deutsch, Español.

Mesh behaviour

Setting	What it does
Bluetooth	Turns the Bluetooth link for the MeshCore phone app on or off.
Bluetooth PIN	The pairing PIN, random for each device. Changing it unpairs every phone.
Advert (Cardputer)	How often an advert is sent automatically: Off, 3, 6, 12 (default) or 24 hours. On the T-Deck this is in the Advertise app.
Telemetry	Whether others may ask for your battery, temperature and (optionally) position.
GPS	Off, Intermittent or On.
Auto Add Contacts	Save nodes to Contacts / Repeaters automatically as their adverts arrive.
DM Retries	How many times an unconfirmed DM is resent along its route, then as a flood. Default 5 / 3.
Region Scope	Your default scope and 9 saved regions. See Regions and scopes.
Location Sharing	Put your position in your adverts.

Device

Setting	What it does
Keyboard Light (T-Deck)	Brightness, Off, or Auto (lit between set times and during Night Mode).
Keyboard Layout	English, French AZERTY or German QWERTZ.
Serial Output	The USB serial log on or off.
Date / Time	Set the clock by hand (it is set from GPS when there's a fix).
Timezone	Hours from UTC.
Firmware Update	Shows the installed version. Updates are flashed over USB (see Installing Saitama).
Save Messages	Save conversations to the SD card (default on).
Show Hops / **Show RSSI**	Show hop count and signal strength on received messages.
Backup & Restore	Write contacts, repeaters and settings to the SD card, or read them back.
Storage Password	The password that encrypts secrets on the SD card. Shows **Default** in red until changed. See Backup and security.
Return to Launcher	Back to the launcher.

Terminal commands

The **Terminal** app takes typed commands. The same commands work over the USB serial port (115200 baud), where the prompt is `OPS>`. Type `help` for the list.

Received messages also appear in the Terminal, as do replies from repeaters.

Device

Command	What it does
`help`	List the commands.
`get radio`	Show the radio settings.
`set name <name>`	Set the device name.
`set lat <deg>` / `set lon <deg>`	Set your position by hand.
`set freq`, `sf`, `bw`, `tx`, `cr`, `af` + value	Set a radio parameter.
`clock [analog]`	Show the time and start the clock screensaver.
`time <epoch-seconds>`	Set the clock.
`memory`	Free memory.
`battery`	Battery voltage and level.
`reboot`	Restart.
`shutdown`	Power off (T-Deck: click the trackball to turn it back on).
`gps [on|off|intermittent]`	Set the GPS mode, or show its status.
`sd`	SD card status. Also `sd mount`, `sd unmount`, `sd restore`, `sd ls [path]`.
`identity show`	Show your node's public key.
`identity restore`	Restore the identity from the SD card backup.
`screenshot [name]`	Save a screenshot to the SD card.
`touch-cali`	Calibrate the touchscreen (T-Deck). `touch-cali reset` undoes it.
`saitama-scr`	Start the Saitama screensaver (T-Deck).

Messages and contacts

Command	What it does
`card`	Print your business card, to share.
`import <card>`	Add a contact from someone's business card.
`list [n]`	List saved contacts (the last n).
`messages <n|all>` / `clearmessages`	Show or clear the message log.
`find <id>`	Look up a contact by ID.

Command	What it does
`to <name|prefix>` then `send <text>`	Send a DM.
`public <text>`	Post on the Public channel.
`ch3 <text>` … `ch5 <text>`	Post on channel slot 3, 4 or 5.
`channels`	List the channels.
`channel <1–10>`	Show one channel, including its key in base64.
`setchannel <1–10> #<name> [psk]`	Set up a channel.
`clearcontacts` / `contacts delete <prefix>`	Remove contacts.
`advert [flood]`	Send an advert now.
`adverts`	Turn on or off showing adverts in the Terminal as they arrive.
`quiet`	Turn new-message popups off or on.

Routes and scopes

Command	What it does
`trace <path>`	Trace a route (T-Deck).
`getpath <hex>` / `setpath <hex> <path>`	Show or set the route to a node.
`resetpath <hex>` / `reset path`	Forget a route.
`scope` / `scope #<region>` / `scope clear`	Show, set or clear the default region scope.
`pathsize [1|2|3]`	Routing hash size in bytes.

Repeaters

Command	What it does
`repeaters [secs]`	Ask all repeaters for their status.
`replist` / `clearrep`	List or clear saved repeaters.
`repeateradmin <hex> [pass]`	Log in to a repeater as admin.
`repadmin [<hex>] <cmd>`	Send an admin command (log in first). Replies appear in the Terminal.

`<hex>` is the start of the node's public key, as shown in Contacts or Repeaters.

Backup and security

What's on the SD card

Saitama keeps a backup of everything important in the `/ops/` folder on the SD card, so a reflash or a new device picks up where you left off:

File	Holds
`/ops/settings.json`	Your settings. Channel keys are stored encrypted.
`/ops/contacts.json`	Saved contacts.
`/ops/repeaters.json`	Saved repeaters, with remembered admin passwords encrypted.
`/ops/identity.enc`	Your node identity (its private key), encrypted.
`/ops/advert_config.json`	The automatic advert interval.
`/ops/msgs/`	Message history, one file per channel or DM.

These are written automatically as things change. **Settings > Backup & Restore** writes or reads contacts, repeaters and settings on demand.

The files are the same on the T-Deck and the Cardputer, so you can move a card between them. The Cardputer holds fewer contacts (100) and repeaters (60) than the T-Deck (250 each).

The storage password

Secrets on the card (channel keys, remembered repeater passwords and the identity backup) are encrypted with AES-256-GCM using your **storage password**.

- It starts as `changeme1`, which is published. **Change it** in **Settings > Storage Password**: until you do, the row shows **Default** in red, and anyone with your SD card can read those secrets.
- The password itself is kept on the device, never on the card.
- **Remember it.** On a new or fully reflashed device, the boot screen asks for it to unlock your identity backup. If you can't give it, the device runs on a temporary identity until you do.

Your identity

Your identity is a key pair. The public half is your address on the mesh; the private half proves messages are from you. It's kept on the device and backed up, encrypted, to `/ops/identity.enc`.

- **Keep the backup.** If you lose the identity, you become a new node: your contacts must add you again, and DMs to the old one can't be read.
- **Settings > Generate Identity** deliberately makes a new one.
- **Terminal:** `identity show` shows your public key; `identity restore` reloads the backup.

Message security

- All mesh traffic is encrypted by MeshCore.

- **Direct messages are verified.** Only you and the recipient can derive the key, so a DM can't be forged in a contact's name. A green padlock (by the sender on the T-Deck, in the title bar on the Cardputer) shows a verified DM. See Chat and channels.
- **Channel messages are not authenticated.** Anyone who has the channel key can post under any name. Treat names on channels as a claim, not proof.
- **Public** uses a key every MeshCore node has. Don't post anything private there.

Bluetooth

The MeshCore phone app can connect over Bluetooth (**Settings > Bluetooth**). The pairing PIN (**Settings > Bluetooth PIN**) is random for each device. Changing it unpairs every phone.

Reporting a security problem

Report it privately to the maintainers, not in a public issue. See docs/SECURITY.md.

Troubleshooting

I can't hear anyone / nobody hears me

- **Radio preset:** everyone on your mesh must use the same preset. Check **Settings > Radio** (or `get radio` in the Terminal) against what your local mesh uses.
- **Antenna:** check it's fitted and screwed in.
- **Region scope:** if local repeaters only forward scoped traffic, set your scope. See Regions and scopes.
- **Finder:** shows which nodes hear you directly. If it finds none, move somewhere higher or more open.
- **Signal:** if the packets-received count isn't rising, the radio is hearing nothing on this preset.

I can't send someone a DM

You need their advert first, and they need yours. Send an advert (T-Deck: **Advertise** app; Cardputer: `advert` in the Terminal), ask them to do the same, and check they appear in **Heard**. Then save them as a contact.

If DMs to a known contact keep failing, the saved route may be out of date: in Contacts choose **Reset Path** and try again.

Messages show as "?" or empty

The device was short of memory when the message arrived. This was much improved in 1.6; if you still see it, reboot, and report it with the serial log.

The Cardputer says memory is low

The Cardputer has no PSRAM, so its memory is tight. When it runs low, Saitama closes the screens you aren't using and shows a notice. Go back to the launcher and reopen the app. Type `mem` in the Terminal to see the free memory.

The device restarted by itself

Connect a USB serial monitor (115200 baud) and look at the first lines after it restarts: `Last reset:` gives the reason (for example a brownout, which points to a weak battery or a very loud speaker volume).

The touchscreen taps in the wrong place (T-Deck)

Type `touch-cali` in the Terminal and tap each crosshair.

The boot screen asks for a password

Your identity backup on the SD card is locked with a storage password that's different from the one on this device (common after a full reflash). Type the storage password you set before. See Backup and security.

The SD card isn't found

Use a FAT32-formatted card. Type `sd` in the Terminal to see its status, and `sd mount` to retry.

Getting help

Open an issue on GitHub with your device, the Saitama version (shown on the splash screen and in **Settings > Firmware Update**), what you did and what happened. A USB serial log helps a lot.

Glossary

The words you'll meet in Saitama and MeshCore, in plain English.

Word	What it means
Mesh	A network made of everyone's radios, passing messages to each other with no phone mast or internet.
MeshCore	The free, open-source system that Saitama uses to run the mesh: how messages are routed, encrypted and passed on.
LoRa	"Long Range": the type of radio used. Slow, but it goes a long way on very little battery.
Node	Any device on the mesh: your handheld, a friend's, or a repeater.
Companion	A device a person carries and chats on, like yours. Shown as **C** in Heard.
Repeater	An always-on radio, usually somewhere high, that passes messages on to extend the mesh. Shown as **R** in Heard.
Hop	One step from one radio to the next. A message that went through two repeaters took 3 hops.
Advert	A short "I'm here" announcement your device sends out. People can only message you once they've heard it.
Flood	Sending a message out to everyone, with every repeater passing it on, to find someone when the route isn't known yet.
Path / route	The list of repeaters a message goes through to reach someone. Once known, messages go that way instead of flooding.
Channel	A group chat. Everyone with the channel's key can read and post.
Public	The channel every MeshCore device has. Anyone can read it.
Hashtag channel	A channel whose name starts with `#`. Its key is made from the name, so anyone who types the same name joins it.
DM (direct message)	A private message to one person. Only the two of you can read it.
Verified / green padlock	A DM that's been checked as really coming from that contact. It can't be faked without their secret key. Channel messages can't be verified.
Key / PSK	The secret that locks a channel. Share it only with people you want in the channel.
Identity	Your device's unique key pair: its address on the mesh, and its proof that messages are really from you.
Region / scope	A tag such as `#london` on a message. Some repeaters only pass on messages with their region's tag.
RSSI	How strong a signal was. Closer to 0 is stronger: –60 is very strong, –120 is very weak.
SNR	How clear a signal was above the background noise. Higher is better.
Noise floor	How much background radio noise there is. Lower is better.
Airtime	How long your radio has spent transmitting. In Europe the law limits how much of the time you may transmit.

Word	What it means
Preset	A ready-made set of radio settings for a country or mesh. Everyone nearby must use the same one.
Firmware	The software that runs on the device itself. Saitama is firmware.
Flash / flashing	Installing firmware onto the device from a computer.
SD card	The memory card that stores your backups, message history and maps.
Storage password	The password that locks the secrets saved on your SD card. Change it from the default.